Preface

Ten years ago, the first edition of *Site Reliability Engineering* set out to explain what Site Reliability Engineering (SRE) is and how Site Reliability Engineers (SREs) do it at Google. In the subsequent years, the role has become widespread beyond Google, and the approach we described has become broadly accepted in the industry. Meanwhile, the technology underpinning running large-scale distributed computing systems has become much more standardized and commoditized: unless you have very specific needs, you're unlikely to build your own load balancer or your own distributed consensus system with Paxos—you're probably either buying it as a service from a cloud provider or running an instance of open source software.

Furthermore, as we write this, AI is in the process of radically accelerating change in our field, as well as many others. We have no crystal ball to forecast what the future of running production environments will be like in another 10 years—or even, quite possibly, 10 *months*.[1]

But what hasn't changed? The fundamentals: SREs continue to be *engineers*, focusing on system *reliability*, for large-scale distributed computing *services*.

All of those trends, combined with what we know has stayed the same, have caused us to shift the focus of this edition toward the things we didn't address in much detail the first time around—things that support well-functioning teams: the sociotechnical aspects of SRE. As a result, you'll see more on understanding the value of what we do and how to provide that value and represent it within an organization—which was important 10 years ago but is even more important now. Something we also learned since the first edition is just how widespread this work—managing the whole lifecycle of software—is, and by how many different names it is called. We wrote in *The Site Reliability Workbook* about the relationship between DevOps and SRE, but platform engineering has emerged since, and staff engineers often have that combination of deep expertise and holistic responsibility that is the hallmark of SRE. Regardless of what the role is called, though, we hope the material in the book will be of relevance to those who care about production.

In the first edition, we used the metaphor of the work before and after the birth of a system being akin to labor and parenting. We believe this analogy still holds, especially in

our current situation (see Chapter 19), where birthing a system has gotten easier but it's not yet clear that parenting or housework has. So much of software engineering is the question of what you can successfully ignore, and the fact that there *is* a whole lifecycle cannot be ignored.

However, another concept from the software engineering discourse has gained prominence in the decade since our first edition: technical debt. This is the term we use to capture the uneasy scenario we arrive at unintentionally when we have built the software on top of itself too much, and everything becomes difficult. Abstractions are stretched to their breaking point; pushing to production feels like trying to hammer a cube through a round hole, where the hammer occasionally bounces off and hits you in the knee. Eventually, you just want to stop and break apart the blocks of what you've built.

We draw a comparison here to technical debt because, despite its ubiquity in software systems, it is often dismissed as only a source of developer sadness. But for those of us who care about production, technical debt is a leading indicator of something important: unreliability.

In the financial world, debt is not inherently bad. It is a leverage tool. You borrow capital to accelerate growth and invest in the future, accepting that you will pay it back with interest. In software, we do the same. We borrow from our future stability to ship a feature today, trading perhaps a robust architecture or the comprehensive testing that would have required more time and effort for speed to market.

But software debt, like financial debt, always demands repayment, with interest.

In production, that interest is paid in complexity, friction, and fragile systems. Every shortcut shrinks our margin for error. If we ignore the debt, the interest compounds silently. Eventually, if left unchecked, we face reliability bankruptcy. This isn't just a slowdown in feature delivery; it is the outage that destroys user trust, or the codebase that's so brittle that no one wants to work on it, much less deploy it.

SRE is not a crusade for zero technical debt—a state that is both economically foolish and practically impossible for any system that is actually useful. Instead, SRE is the discipline of managing this risk portfolio. We are both engineers and production advisors, helping the organization decide when to borrow safely and when we must pay down the principal according to our agreed-upon risk ceilings, temporarily halting feature development to invest in repayment.

This edition is about how we manage that balance. Many of the systems we run and the approaches we take have changed in the 10 years since the first edition, but this hasn't:

reliability is not a static state to be achieved and forgotten. It is an ongoing negotiation between complexity and business needs. The chapters that follow explore the tools, culture, and practices we use at Google to successfully navigate that complexity to provide reliable services.

HOW TO READ THIS BOOK

Like the first edition, this book is a series of essays written by members and alumni of Google's Site Reliability Engineering organization. This time, we have expanded our view with authors from our Google partner teams who work with Google Cloud customers, and even to case studies from Disney, Etsy, Microsoft, and Spotify. It's much more like conference proceedings than it is like a standard book by a single author or a small number of authors. Each chapter is intended to be read as part of a coherent whole, but a good deal can be gained by focusing on whatever subject particularly interests you. Where other chapters or sections support or inform the text, we reference them so you can explore accordingly.

While the first edition of this book covered the foundational principles and practices of SRE, this edition delves into how these concepts and practices have developed and matured over the past decade. Additionally, we tackle some new topics: STPA and CAST (Chapter 12), and how AI is changing the SRE profession (Chapters 19, 20, and 22). We chose not to spend much time on topics that are evergreen and were well covered in the previous edition. As a result, you'll see that the chapters don't map one to one from the first edition to this one. It's helpful to think of this work as a supplement to that book, rather than a wholesale replacement.

You don't need to read the book in any particular order, though we'd suggest at least starting with Chapters 1, 2, and 4. Reading cover to cover is, of course, also useful and possible. The chapters are grouped thematically into six parts: Part I, "SRE in Context"; Part II, "Sociotechnical Foundations"; Part III, "Situational Awareness and Continuous Preparedness"; Part IV, "Infrastructure Design"; Part V, "Resilient Products and Emergent Reliability"; and Part VI, "Outlook". Each part has a small introduction that highlights what the individual chapters are about.

We hope this book will be at least as useful and interesting to you as putting it together was for us.

— The Editors

Guidance for Specific Groups of Readers

As we approached this book, it was helpful for us to think of broad categories of readers to address. We identified the following three groups to guide our writing:

Aspiring SREs

Individuals who are new to core SRE concepts, such as recent graduates or those transitioning into SRE from another discipline

Practitioners

Seasoned technical engineers who are actively working hands-on in the SRE space

Leaders

Executives, managers, and C-suite individuals who need to understand the strategic, organizational, and business value of SRE

While no grouping or reading guide is perfect or perfectly comprehensive, here are some groupings of content you might find particularly helpful based upon which of these categories you fall into.

Aspiring SREs: Context and Overviews

We recommend the following resources for aspiring SREs:

The first edition of this book

Covers the foundational principles and practices of SRE.

The Foreword

Sets the historical stage for this book, reflecting on the last decade of SRE and introducing overarching concepts like "reliability by default."

Introductions to each of the six parts

Help orient newcomers to the themes, definitions, and upcoming technical concepts of each major section of the book.

Chapter 1, "Introduction: Modeling What We Do and Why"

Serves as a structural roadmap to orient the reader to the landscape of SRE concepts covered in this book.

Chapter 3, "Trends in Production"

Explains how modern production ecosystems have evolved, in addition to the broad industry shifts (CapEx to OpEx, virtual machines to containers, FinOps) that establish the parameters of modern production.

Chapter 10, "Incident Management and On-Call"

Discusses setting up on-call rotations, establishing incident response teams (IRTs), and mitigating live outages.

Enterprise Roadmap to SRE

Provides a step-by-step approach for integrating the SRE function into an organization.

Practitioners: Technical Implementation and Operations

We recommend the following resources for practitioners:

Chapter 1, "Introduction: Modeling What We Do and Why"

Serves as a structural roadmap to orient the reader to the landscape of SRE concepts covered in this book.

Chapter 8, "Service Level Objectives"

Provides practical advice for defining service level indicators (SLIs), setting error budgets, writing alerting rules, and implementing client-side/per-customer monitoring.

Chapter 9, "Observability and Monitoring"

Provides architectural guidance for building observability platforms, using OpenTelemetry, and instrumenting client-side telemetry.

Chapter 10, "Incident Management and On-Call"

Discusses setting up on-call rotations, establishing IRTs, and mitigating live outages.

Chapter 11, "Learning from Incidents"

Outlines the hands-on process of conducting blameless postmortems, layered reviews, and integrating systems thinking into daily operations.

Chapter 12, "Safety Engineering for SRE with STPA and CAST"

Takes a deep dive into applying the STAMP, STPA, and CAST frameworks for proactive hazard analysis and complex root-causing.

Chapter 13, "Automation and Tooling"

Provides tactical frameworks for when and how to write automation, evaluating technical debt, and avoiding the "ironies of automation."

Chapter 14, "Testing"

Provides a roadmap for transforming testing from a mere pre-deployment gate into a proactive operational strategy, leveraging advanced techniques like testing in production, fault injection, and formal methods.

Chapter 15, "Capacity Planning"

Offers technical guidance on load shedding, managing failure domains, resource pooling, and the mathematics of capacity efficiency.

Chapter 16, "Data Management"

Details specific strategies for data replication, differential versus incremental backups, checksumming, and restoring corrupted data.

Chapter 17, "Designing for Resilience"

Provides a technical view on implementing architectural failure domains, graceful degradation, adaptive throttling, and fallback mechanisms.

Chapter 18, "Platform Engineering in SRE"

Outlines how platform engineering in SRE can reduce cognitive load and improve reliability by shifting infrastructure complexity into "golden paths," standardized policies, and curated self-service solutions.

Chapter 20, "How SRE Supports AI"

Focuses on the technical challenges of managing ML/AI workloads, such as TPU capacity planning, ML goodput, and building model qualification pipelines.

Chapter 21, "How AI Supports SRE"

Provides a framework for transitioning from deterministic automation to probabilistic "SRE AI," while ensuring autonomous agents safely manage large-scale

system complexity.

Chapter 22, "SRE in Diverse Environments"

Offers pragmatic guides for solo SREs at startups and engineers navigating constrained, legacy, or heavily regulated environments.

Leaders: Strategy, Culture, and Organization

We recommend the following resources for leaders:

Chapter 1, "Introduction: Modeling What We Do and Why"

Serves as a structural roadmap to orient the reader to the landscape of SRE concepts covered in this book.

Chapter 2, "The Value of Reliability"

Focuses on the business case for SRE, covering sublinear scaling, FinOps, and the financial/opportunity costs of unreliable systems.

Chapter 4, "The Cultural Context of SRE"

Discusses some of the research behind adoption of SRE practices, in addition to psychological safety and how to foster a generative culture of learning and continuous improvement.

Chapter 5, "SRE Hiring, Training, and Career Management"

Provides guidance for recruiting, creating hiring pipelines, building onboarding programs, and managing retention.

Chapter 6, "Organizational Structures for SRE"

Explores the strategic design of teams (centralized versus decentralized), maturity matrices, and how scorecards can factor into governing SRE performance.

Chapter 7, "SRE Engagement Models"

Details the strategic frameworks for how SRE teams interface with product development and how to negotiate those operational boundaries.

Chapter 19, "SRE at the Tipping Point"

Reflects on how AI is compressing the traditional software development lifecycle (SDLC) and what that means.

Chapter 21, "How AI Supports SRE"

Presents a strategic vision for integrating AI into workflows, emphasizing maturity models, governance, and the organizational shift required for AI adoption.

Chapter 23, "Future of SRE"

Examines industry-wide strategic shifts, such as "shifting down" to platforms, treating risk as a leading indicator, and the evolving role of SREs in the era of AI.

We've put thought into learning paths for each of these audiences—please visit *https://sre.google* to see them.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions

`Constant width`

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords

---

TIP

This element signifies a tip or suggestion.

NOTE

This element signifies a general note.

WARNING

This element indicates a warning or caution.

Using Code Examples

Supplemental material (code examples, exercises, etc.) is available for download at *https://sre.google*. This book is also available under the Creative Commons Attribution-Noncommercial-No Derivative Works 4.0 International Public License (CC BY-NC-ND 4.0).

If you have a technical question or a problem using the code examples, please email *support@oreilly.com*.

This book is here to help you get your job done. In general, if example code is offered with this book, you may use it in your programs and documentation. You do not need to contact us for permission unless you're reproducing a significant portion of the code. For example, writing a program that uses several chunks of code from this book does not require permission. Selling or distributing examples from O'Reilly books does require permission. Answering a question by citing this book and quoting example code does not require permission. Incorporating a significant amount of example code from this book into your product's documentation does require permission.

We appreciate, but generally do not require, attribution. An attribution usually includes the title, author, publisher, and ISBN. For example: "*Site Reliability Engineering*, 2nd ed., edited by Betsy Beyer, Chris Jones, Christof Leng, David Huska, Jennifer Petoff, and Niall Richard Murphy (O'Reilly). Copyright 2026 Google LLC and Niall Murphy, 979-8-341-60768-2."

If you feel your use of code examples falls outside fair use or the permission given above, feel free to contact us at *permissions@oreilly.com*.

O'Reilly Online Learning

NOTE

For more than 40 years, O'Reilly Media has provided technology and business training, knowledge, and insight to help companies succeed.

Our unique network of experts and innovators share their knowledge and expertise through books, articles, and our online learning platform. O'Reilly's online learning platform gives you on-demand access to live training courses, in-depth learning paths, inter-

active coding environments, and a vast collection of text and video from O'Reilly and 200+ other publishers. For more information, visit *https://oreilly.com*.

How to Contact Us

Please address comments and questions concerning this book to the publisher:

O'Reilly Media, Inc.

141 Stony Circle, Suite 195

Santa Rosa, CA 95401

800-889-8969 (in the United States or Canada)

707-827-7019 (international or local)

707-829-0104 (fax)

support@oreilly.com

https://oreilly.com/about/contact.html

We have a web page for this book, where we list errata and any additional information. You can access this page at *https://oreil.ly/SRE-2e*.

For news and information about our books and courses, visit *https://oreilly.com*.

Find us on LinkedIn: *https://linkedin.com/company/oreilly*.

Watch us on YouTube: *https://youtube.com/oreillymedia*.

Acknowledgments

This book would not have been possible without the tireless efforts of our authors, technical writers, and reviewers. Matt Zelesko, Kesh Subramanian, and Jenn Petoff were this book's sponsors within Google; their belief in this project and sharing what we've learned about running large-scale services was essential to making this book happen.

While the authors and technical writers are specifically acknowledged in each chapter, we'd like to recognize those who contributed to each chapter by providing thoughtful input, discussion, and review. In chapter order, they are:

- Chapter 1: JC van Winkel
- Chapter 2: Pereira Braga
- Chapter 3: Alex Matey, Ben Appleton, JC van Winkel, Steve McGhee
- Chapter 4: James Brookbank, Marcus Mitchell, Preston Yoshioka, Trisha Weir
- Chapter 5: Andrew Harvey, Angela Karam, Brad Lipinski, Daniel Crawford, Ed Doyle, JC van Winkel, Jess Der, Josh Kenitzer, Laura Baum, Mallory Rivera, Pim van Pelt, Preston Yoshioka, Roland Hess, Tom Cranitch
- Chapter 6: Hannes Eder
- Chapter 7: Andrew Calman, Sanjay Sha
- Chapter 8: Roman Seibel, Stepan Davidovic
- Chapter 9: Alan Antonuk, Andrei Chirila, Brice Arnould, Jake Finnegan, James Maffey, Jeffrey Borwey, Jose Andrade, Julius Plenz, Mary Koes, Peter Balsiger, Robert Lehmann, Rowan Kloti, Sidnei Silva, Stepan Davidovic, Steve McGhee, Sujay Solomon, Summit Tuladhar, Tyler Sanderson
- Chapter 10: Nick Soda, Trevor Mattson-Hamilton
- Chapter 11: Nick Soda, Trevor Mattson-Hamilton
- Chapter 12: David Ferguson, Theo Klein
- Chapter 13: Jessica Hainey, Stephen Ryan
- Chapter 14: Adriana Maggiore, Alex Perry, Allison Morris, Axel Freyn, Barclay Osborn, Boris Bokowski, Brian Roddy, Chris Wasik, Corinna Cortes, Eric Miller, Hariharasudhan Reghuraman, Jeff Mogul, Kieran Broadfoot, Rich Alimi, Sean Finney, Steffen Smolka, Suruchi Pundir, Thyla van der Merwe, Timon Van Overveldt, Vitaliy Shipitsyn
- Chapter 15: Andrew Calman, Des Keane, Tom Kennedy
- Chapter 16: Andrew Calman, Catalin Nicutar
- Chapter 17: Kieran Broadfoot
- Chapter 18: James Brookbank, Salim Virji
- Chapter 20: Carl Evankovich, Jake Prasad
- Chapter 21: Ioannis Papapanagiotou

We very much appreciate the thoughtful and in-depth feedback that we received from external reviewers: Daria Barteneva, David Blank-Edelman, Matt Brown, Shilpa Daga, Chris Devers, Noah Gift, David Hixson, Laura Nolan, John O'Connor, Matt Olson, Lipi Deepaakshi Patnaik, J. Paul Reed, Vishwesh Ravi Shrimali, Andrea Spadaccini, Todd Underwood, and Jesse van Herk. Thank you also to Randall Munroe for letting us use his xkcd comic "Is It Worth the Time?".

We would like to extend special thanks to Shylaja Nukala, who generously gave of the time of her technical writers and supported their necessary and valued efforts whole-heartedly. Henry Faulkner has been instrumental in making all the logistics of our work-

ing sessions magically "just work." A big thanks to Amanda Corey for her timely and thoughtful review of large swaths of this book.

The editors would also like to personally thank the following people:

Betsy Beyer: To Jenn and Kesh, for their incredible mentorship and on-the-ground support in getting this project over the line. To Polly, Trudy, Gregg, and Mary for always inspiring me to put my best foot forward. To Elliott, the Beyers, the Simpsons and Todds, and my grandparents for existential happiness.

Chris Jones: To everyone I've worked with and learnt from over the years. To Betsy, Christof, David, Jenn, and Niall, who made this project a joy. And to Sally, Sprout, and my family, for everything.

Christof Leng: To my wife Vera and our sons Samson, Adam, and Joshua. You are my life. To my mother and siblings for helping me become who I am. To all the great people at Google I have worked with and who helped me learn and grow. To Betsy, Chris, David, Jenn, and Niall for letting me tag along. It's an incredible honor.

David Huska: To my parents. To all the engineers whose imagination and reason I stand on every day. To my coauthors, whose experience and skill made this book a reality. And to anyone who's ever learned they were on-call for a service by receiving a page for it.

Jennifer Petoff: To Betsy—who knew that the SRE Book would be the start of a decade of collaboration and friendship? To Niall, for bringing me on board to program manage his passion project to write a book about SRE—which became the coolest thing that I worked on in almost two decades at Google. So grateful that we got to get the band back together (and even added a couple of new members!) to write this second edition. To my husband Scott, for being so supportive of all my SRE-fueled adventures and for coming along for the ride.

Niall Richard Murphy: To Léan, Oisín, and Fiachra, the reasons I do all this. To Kay, forever and endlessly. To the Grays (yes, all of you). To Paul, in a sun-dappled glade. To Todd, without artifice. To Cian, amiably and lovably intransigent. To this weird thing, whatever it is, that I do, that I'd previously never heard of and I'm still not quite sure I understand, but that remains fascinating 20 years on. To the elegiac future.

1 But see Chapter 19 for some thoughts!

Previous chapter

< Foreword

Next chapter

I. SRE in Context >